

1. Administrative & Study Identification

Full Study Title: A Translational Study to Describe Asthma Patient Clinical Characteristics, Treatment Patterns, Biomarkers and to Identify Phenotypes and Endotypes associated with Differential Outcomes that may Support Future Development of Personalised Treatment Strategies in Chinese Population **Short Title/Acronym:** Asthma Biomarkers and Endotypes Translational Study **Protocol Number:** D2287R00186 **NCT Number:** NCT06419413 **Sponsor Name:** AstraZeneca **Investigational Product:** N/A (Observational/Translational Study) **Phase of Development:** N/A **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To describe clinical characteristics, treatment patterns, and biomarkers, and to identify phenotypes and endotypes associated with differential outcomes in the Chinese asthma population. **Secondary Objectives:** To establish statistical clustering of key biomarkers across different asthma severities and assess the statistical association between specific biomarkers and clinical disease outcomes.

2.2 Background & Rationale To address the gap in understanding asthma phenotypes and endotypes specifically within the Chinese population. The study aims to map clinical characteristics and biomarker levels across healthy participants, mild asthmatics, moderate-to-severe asthmatics, and those with exacerbations, ultimately supporting the future development of personalized and targeted treatment strategies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational / Translational **Control Type:** Healthy Volunteer Cohort (Cohort A) compared against disease severity cohorts **Blinding:** Open-label **Randomization:** N/A (Cohort assignment based on diagnosis and severity)

3.2 Planned Enrollment & Duration Target \$N\$: Up to 355 participants (approx. 40 healthy, 40 mild asthma, 160 moderate-to-severe asthma, and 75 with exacerbations) **Number of Sites:** 4 study centers in China (Chengdu, Taiyuan, Wuhan, Xi'an) **Estimated Study Start:** 08-Jan-2024 **Estimated Primary Completion:** 08-Oct-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 75 years. 2. Acceptable FEV1 measurements according to ATS and ERS guidelines. 3. For all Asthma Cohorts: Physician-diagnosed asthma ≥ 3 months prior to the screening visit.

4.2 Exclusion Criteria 1. Pregnant at the time of an assessment. 2. History of alcohol or drug abuse within the past year, or altered mental status at the time of informed consent. 3. Receipt of marketed or investigational biologic(s) within 3 months or 5 half-lives prior to visit 1. 4. Receipt of LTRAs or 5-lipoxygenase (5-LO) inhibitors (e.g., zileuton and montelukast) within 1 month or 5 half-lives prior to baseline.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: N/A (No investigational product administered). Patients will remain on their standard background therapy (e.g., low-dose ICS-formoterol, SABA, LABA) according to their assigned cohort. **Route of Administration:** N/A **Duration of Treatment:** Observational study; duration corresponds to standard clinical follow-up or exacerbation event assessment.

5.2 Concomitant & Prohibited Therapy Permitted: Standard asthma therapies including short-acting β_2 agonists (SABA), long-acting β_2 agonists (LABA), and inhaled corticosteroids (ICS). **Prohibited:** Investigational or marketed biologics (within 3 months), and leukotriene receptor antagonists (LTRAs) or 5-LO inhibitors (within 1 month).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -14 to 0 Informed Consent, Medical History, Spirometry (FEV1), Asthma Diagnosis Verification
Baseline	Day 0	Cohort Assignment (A, B, C, or D), Clinical Characteristics Assessment, Biomarker Sample Collection	
Interim	TBD	Ongoing Observation, Assessment of Exacerbations (for Cohort D)	End of Study TBD Final Data Collection, Biomarker Analysis, Exit Review

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Descriptive summary of demographic, subject characteristics, clinical characteristics, and functional deliverables at baseline by study cohorts.

Measurement Method: Clinical assessment and patient records review.

7.2 Secondary & Exploratory Endpoints * Descriptive summary of biomarker levels at baseline by study cohorts. * Statistical clustering of key biomarkers in the moderate to severe asthma cohort. * Statistical association between the biomarker profiles and clinical disease outcomes in each asthma cohort.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard reporting applicable to observational studies and biomarker sample collection procedures. **Serious Adverse Events (SAEs):** Routine reporting within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal safety monitoring mechanisms managed by the sponsor (AstraZeneca).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All enrolled participants meeting cohort criteria (Healthy, Mild, Moderate-to-Severe, Exacerbation). **Sample Size Justification:** $N \leq 355$ is targeted to achieve sufficient statistical power for biomarker clustering, ensuring at least 160 participants in the moderate-to-severe group and specific demographic targets (e.g., ≥ 20 smokers/ex-smokers and ≥ 40 males ≥ 40 years old in Cohort A). **Handling of Missing Data:** Standard descriptive statistics with data imputation techniques applied as necessary for multivariate clustering algorithms.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine source data verification for observational records and sample management. **Audit Trail:** Standardized tracking of biological samples, eCRF locking, and data clarification mechanisms.

11. Study Identifiers & Governance

Primary ID: D2287R00186 **Registry Number:** NCT06419413 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** A Translational Study to Describe Clinical Characteristics, Biomarkers and to Identify Phenotypes and Endotypes Associated With Differential Outcomes in Chinese Population **Official Regulatory Title:** A Translational Study to Describe Asthma Patient Clinical Characteristics,

Treatment Patterns, Biomarkers and to Identify Phenotypes and Endotypes associated with Differential Outcomes that may Support Future Development of Personalised Treatment Strategies in Chinese Population

12. Clinical Framework (Asthma Specific)

Primary Condition: Asthma **Diagnosis Threshold:** Physician-diagnosed asthma \$\\ge\$ 3\$ months prior to screening with acceptable FEV1 **Medical Methodology:** Biomarker analysis and phenotype/endotype statistical clustering **Optimization Target:** Future development of personalized treatment strategies based on local population endotypes

13. Study Procedures & Methodology

Management Pathway: Standard of care for asthma (Cohorts B-D) **Education Protocol (HCP):** Protocol specific training regarding sample collection and accurate cohort phenotyping **Education Protocol (Patient):** Informed consent and overview of sample collection procedures **Quality Audit Mechanism:** Centralized laboratory auditing for biomarker analysis and respiratory function test standardization

14. Observation & Follow-Up Schedule

Total Duration: Cross-sectional baseline with longitudinal exacerbation follow-up **Onsite Visit Cadence:** Baseline visit with specific visits corresponding to protocol-defined exacerbations (Cohort D) **Remote Monitoring Frequency:** As required per standard observational tracking **Checklist Review Interval:** Routine eCRF review during the enrollment phase

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	02-Apr-2026				
Clinical Operations Lead	[Name]	_____	02-Apr-2026		Lead		
Statistician	[Name]	_____	02-Apr-2026				